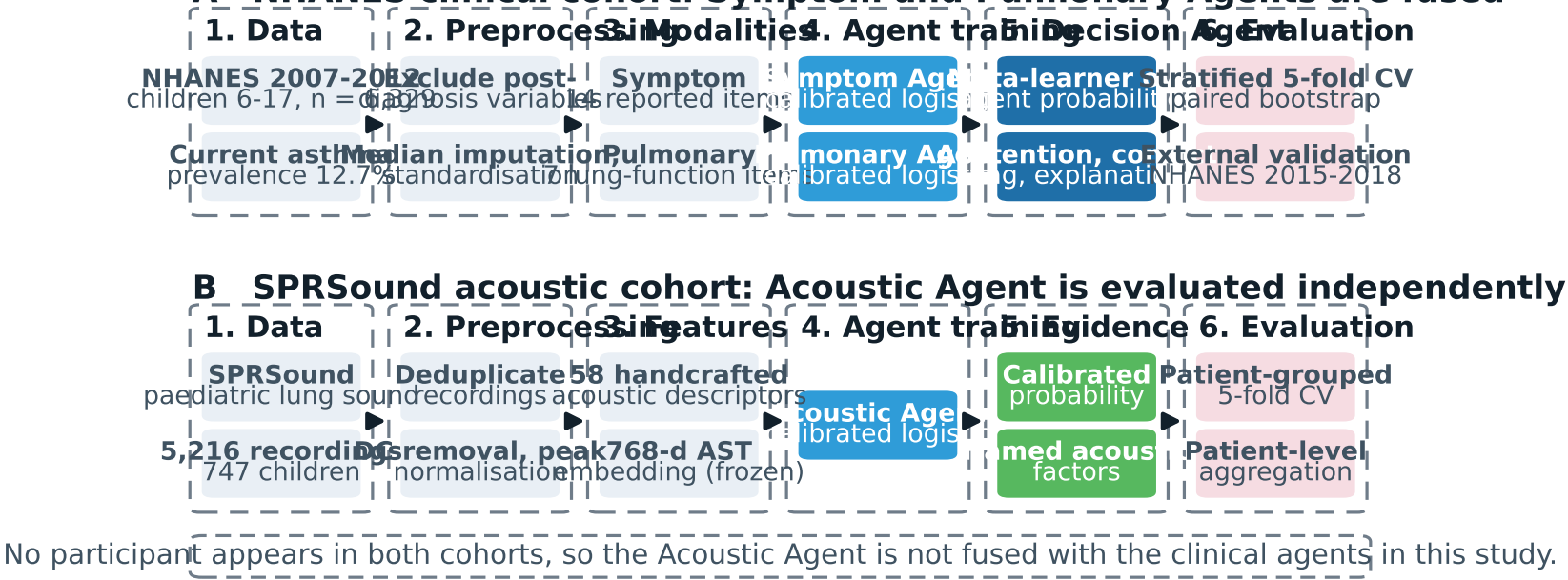
A NHANES clinical cohort: Symptom and Pulmonary Agents are fused
1. Data
2. Preprocessing
3. Modalities
4. Agent training
5. Decision Agent
6. Evaluation
NHANES 2007-2012
children 6-17, n = 6,329
Exclude post-
Symptom
mptom Age
ca-learner
Stratified 5-fold CV
Current asthma
prevalence 12.7%
Median imputation,
standardisation
Pulmonary
lung-function item
monary Ag
tention, co
External validation
NHANES 2015-2018
B SPRSound acoustic cohort: Acoustic Agent is evaluated independently
1. Data
2. Preprocessing
3. Features
4. Agent training
5. Evidence
6. Evaluation
SPRSound
paediatric lung sound
Deduplicate
recordings
58 handcrafted
acoustic descriptors
coustic Age
Calibrated
probability
Patient-grouped
5-fold CV
5,216 recordings
747 children
DC removal, peak
normalisation
768-d AST
embedding (frozen)
med acous
factors
Patient-level
aggregation
No participant appears in both cohorts, so the Acoustic Agent is not fused with the clinical agents in this study.